



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
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DSA0404 - FUNDAMENTALS OF DATA SCIENCE
UNIT 3 - ASSESSMENT TOOL 2 [CO3-AT2]

DATA
INTERPRETATION

SUBMITTED BY
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1. SAMPLING DISTRIBUTION REASONING

Question:

A population has mean $\mu = 500$ and SD $\sigma = 80$. A random sample of $n = 64$ is drawn.

- (a) Compute the standard error of the sample mean.
- (b) What is the probability that the sample mean exceeds 520, assuming the sampling distribution is approximately normal?
- (c) If n were increased to 256, how does the standard error change, and what principle explains this?

Solution

(a) Standard Error:

$$SE = \frac{\sigma}{\sqrt{n}} = \frac{80}{\sqrt{64}} = 10 \Rightarrow \boxed{SE = 10}$$

(b) Probability that $\bar{X} > 520$:

$$Z = \frac{520 - 500}{10} \Rightarrow \boxed{Z = 2}$$

From the standard normal table

$$P(Z > 2) = 1 - 0.9972 \Rightarrow \boxed{0.0228}$$

(c) For ($n=256$):

$$SE = \frac{80}{\sqrt{256}} = \frac{80}{16} = \boxed{5}$$

The standard error decreases from 10 to 5. This illustrates the square-root law, where increasing sample size reduces sampling variability.

2. COVARIANCE CALCULATION AND SIGN INTERPRETATION

Question:

Given paired data on advertising spend (X , ₹'000) and sales (Y , ₹ lakh):

X : 10, 20, 30, 40, 50

Y : 15, 18, 25, 30, 42

Calculate $\text{Cov}(X, Y)$. Is the relationship direction consistent with intuition? What does the magnitude alone fail to tell you (that correlation would)?

Solution

$$\bar{X} = \frac{10 + 20 + 30 + 40 + 50}{5} = 30$$

$$\bar{Y} = \frac{15 + 18 + 25 + 30 + 42}{5} = 26$$

X	Y	(X - $\bar{X}$)	(Y - $\bar{Y}$)	Product
10	15	-20	-11	220
20	18	-10	-8	80
30	25	0	-1	0
40	30	10	4	40
50	42	20	16	320

$$\sum (X - \bar{X})(Y - \bar{Y}) = 660$$

Using Population Covariance,

$$\text{Cov}(X, Y) = \frac{660}{5} = 132$$

Interpretation:

Covariance is positive, so advertising spend and sales tend to increase together. This is consistent with intuition.

The magnitude of covariance alone cannot indicate the strength of the relationship, because covariance depends on the units/scales of the variables. Correlation is standardized between -1 and $+1$ and therefore gives a comparable measure of relationship strength.

3. CORRELATION VS CAUSATION TRAP

Question

A city reports that ice cream sales and drowning incidents have a correlation coefficient of $r = 0.85$. A newspaper claims "ice cream causes drowning." Identify the statistical flaw, name the likely confounding variable, and explain what additional evidence (not just correlation) would be needed to claim causation.

Solution

The statistical flaw is assuming correlation implies causation.

A likely confounding variable is temperature/weather:

$$\text{Hot Weather} \rightarrow \begin{cases} \text{Higher ice cream Sales} \\ \text{More swimming} \rightarrow \text{More drowning incidents} \end{cases}$$

Thus, the high correlation does not prove that ice cream causes drowning.

To establish causation, additional evidence such as controlled studies, longitudinal analysis, or statistical analysis controlling for temperature and other confounding factors would be required.

4. EFFECT OF OUTLIERS ON CORRELATION

Question

A dataset of 9 points shows $r = 0.42$ (weak positive). One extreme outlier is removed, and r becomes 0.89. What does this reveal about the fragility of Pearson's correlation? Would Spearman's rank correlation be more robust here? Justify.

Solution

The change from:

$$r = 0.42 \rightarrow r = 0.89$$

shows that Pearson's correlation is highly sensitive to extreme outliers. A single unusual observation can substantially alter the measured strength of a linear relationship.

Yes, Spearman's rank correlation would generally be more robust to outliers because it uses the ranks of the observations rather than their raw magnitudes.

Therefore, when extreme values are present, Spearman's correlation can provide a more stable measure of the monotonic relationship.

5. DATA PREPARATION - MISSING VALUES

Question

A dataset of 200 rows has 15% missing values in the "Income" column, and analysis shows these missing values are more common among unemployed respondents (not random).

(a) Classify this missing data mechanism (MCAR/MAR/MNAR).

(b) Why would mean imputation introduce bias here?

(c) Suggest a better strategy.

Solution

(a) Missing-data mechanism:

MAR (Missing At Random)

The missingness is associated with employment status, an observed variable.

(b) Why mean imputation is biased:

Using the overall mean income ignores the relationship between employment status and income. Since missing values are more common among unemployed respondents, replacing them with the overall mean can overestimate the incomes of the missing group and distort statistical results.

(c) Better strategy:

Use group-wise imputation based on employment status or a model-based/multiple imputation method that incorporates relevant variables.

Group-wise or model-based imputation is preferable to overall mean imputation.